

Mathematical Modeling of Chronic Myelogenous Leukemia (CML)

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Acknowledgments

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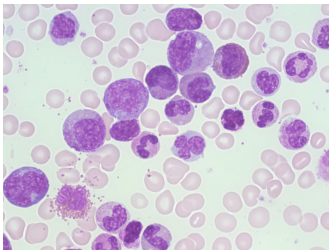
We would like to thank the organizers of JMM 2025 for accepting our abstract for the PME Contributed Session on Research by Undergraduates.

Paper in Review!

Paquin, D., Gross, L., Stewart, A., Thai, G. (2024). *Numerical Analysis of Critical Parameter Values for Remission During Imatinib Treatment of Chronic Myelogenous Leukemia*. Submitted to Mathematical Biosciences and Engineering: Special Issue on Models and Applications of Delay Differential Equations.

Overview of Talk

- 1 Introduction to CML, DDE Model
- 2 Computational Implementation, Motivation for STI
- 3 Threshold Analysis Setup, Case Study



What is CML?

- Chronic myelogenous leukemia (CML) is a cancer of the blood cells and bone marrow.
- CML causes uncontrolled growth of white blood cells, preventing development of healthy white blood cells, red blood cells, and platelets.
- CML is treated with imatinib (marketed as Gleevec).
- Most patients with CML are able to attain some form of remission with prolonged imatinib treatment.
- However, imatinib does not completely eradicate leukemia, and if treatment is stopped, all patients eventually relapse.
- Acquired immune resistance to imatinib is a significant challenge for CML treatment programs.

Modeling Cell Populations with DDEs

$$\begin{array}{ccccccc} \text{Change in} & = & \text{Natural} & - & \text{Natural} & - & \text{Immune} \\ \text{CML} & & \text{Growth} & & \text{Death} & & \text{Response} \\ & & & & & & \text{(T-cells)} \end{array}$$

$$\begin{array}{ccccccc} \text{Change in} & = & \text{Supply} & - & \text{Natural} & - & \text{CML} \\ \text{T-cells} & & \text{Rate} & & \text{Death} & & \text{Interactions} \\ & & & & & & + \text{Stimulated} \\ & & & & & & \text{Divisions} \end{array}$$

Modeling Cell Populations with DDEs

$$\begin{array}{rclclcl}
 \text{Change in CML} & = & \text{Natural Growth} & - & \text{Natural Death} & - & \text{Immune Response (T-cells)} \\
 \frac{dy_1}{dt} & = & a_y y_0 & - & d_1 y_1 & - & q_C p(C, T) y_1
 \end{array}$$

$$\begin{array}{rclclclcl}
 \text{Change in T-cells} & = & \text{Supply Rate} & - & \text{Natural Death} & - & \text{CML Interactions} & + & \text{Stimulated Divisions} \\
 \frac{dT}{dt} & = & s_T & - & d_T & - & p(C, T) C & + & 2^n p(C_{n\tau}, T_{n\tau}) q_T C_{n\tau}
 \end{array}$$

$C(t - n\tau)$
 $T(t - n\tau)$

DDE System for CML Dynamics (Kim *et al*, 2008)

$$\begin{array}{l} \text{Non-} \\ \text{Mutated} \\ \text{Leukemia} \end{array} \left\{ \begin{array}{l} \frac{dy_0}{dt} = (r_y(1 - u) - d_0)y_0 - q_C p(C, T)y_0 \\ \frac{dy_1}{dt} = a_y y_0 - d_1 y_1 - q_C p(C, T)y_1 \\ \frac{dy_2}{dt} = b_y y_1 - d_2 y_2 - q_C p(C, T)y_2 \\ \frac{dy_3}{dt} = c_y y_2 - d_3 y_3 - q_C p(C, T)y_3 \end{array} \right.$$

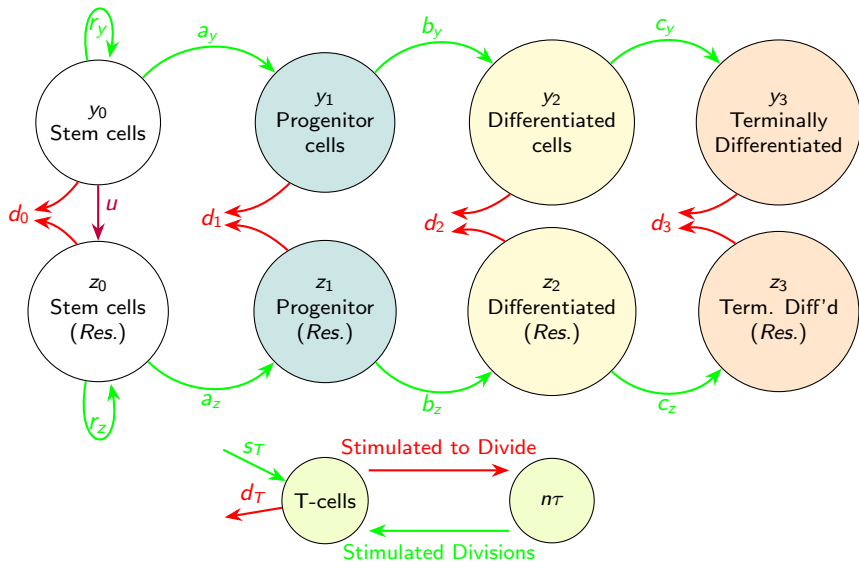
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$$\begin{array}{l} \text{Imatinib-} \\ \text{Resistant} \\ \text{(Rate } u) \end{array} \left\{ \begin{array}{l} \frac{dz_0}{dt} = (r_z - d_0)z_0 + r_y u y_0 - q_C p(C, T)z_0 \\ \vdots \end{array} \right.$$

DDE System for CML Dynamics (Kim *et al*, 2008)

$$\begin{array}{l}
 \text{Non-Mutated Leukemia} \quad \left\{ \begin{array}{l} \frac{dy_0}{dt} = (r_y(1 - u) - d_0)y_0 - q_C p(C, T)y_0 \\ \frac{dy_1}{dt} = a_y y_0 - d_1 y_1 - q_C p(C, T)y_1 \\ \frac{dy_2}{dt} = b_y y_1 - d_2 y_2 - q_C p(C, T)y_2 \\ \frac{dy_3}{dt} = c_y y_2 - d_3 y_3 - q_C p(C, T)y_3 \end{array} \right. \\
 \text{Imatinib-Resistant (Rate } u) \quad \left\{ \begin{array}{l} \frac{dz_0}{dt} = (r_z - d_0)z_0 + r_y u y_0 - q_C p(C, T)z_0 \\ \vdots \end{array} \right. \\
 \text{Changes in T-cell Population} \quad \left\{ \begin{array}{l} \frac{dT}{dt} = s_T - d_T T - p(C, T)C + 2^n p(C_{n\tau}, T_{n\tau})q_T C_{n\tau} \\ C_{n\tau} = C(t - n\tau), \quad T_{n\tau} = T(t - n\tau) \end{array} \right.
 \end{array}$$

Visual Schematic: Kim *et al.* Model



CML and T-Cell Interaction Terms

Total Cancer Population:

$$C(t) = y_0(t) + \dots + y_3(t) + z_0(t) + \dots + z_3(t)$$

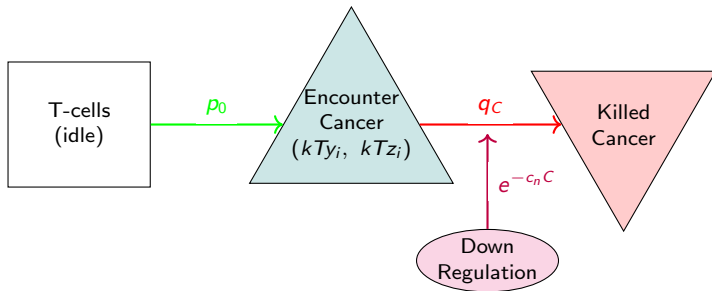
Interactions between Cancer and T-cells:

$$p(C, T) = p_0 e^{-c_n C} k T$$
$$C_{n\tau} = C(t - n\tau); T_{n\tau} = T(t - n\tau)$$

- p_0 : Probability that T-cell engages cancer cell upon interaction
- $e^{-c_n C}$: Immune down-regulation; as cancer increases, ability to fight back decreases.
- k : Mixing coefficient
- T : Concentration of anti-leukemia T-cells

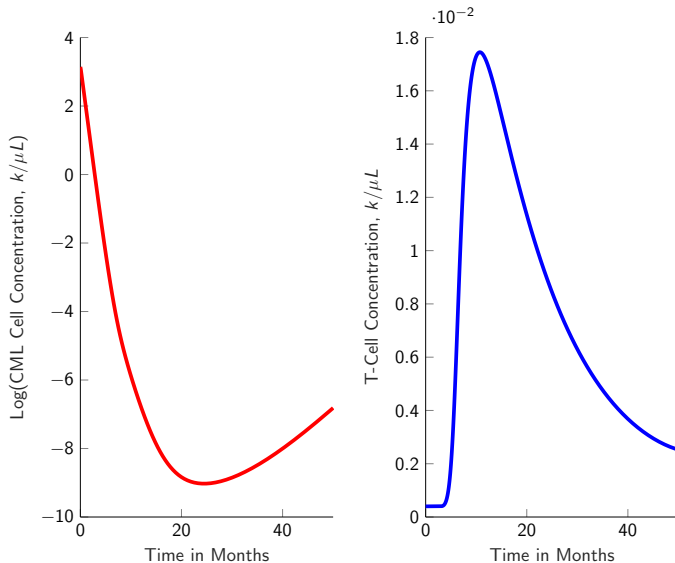
Breaking Down the Anti-Leukemia Immune Response

$$q_C \cdot p(C, T) \cdot y_0 = q_C \cdot p_0 \cdot e^{-c_n C} \cdot kTy_0$$

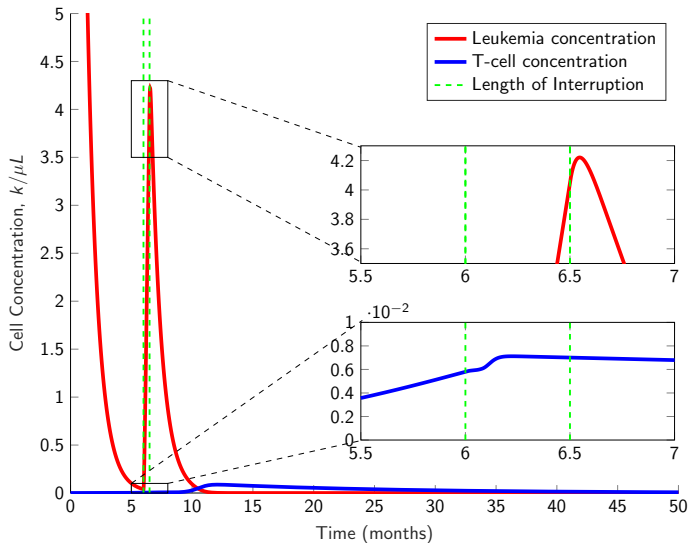


- kTy_0 : **Rate of interaction** between T-cells and cancer cells.
- p_0 : Probability that **T-cell engages** cancer cell upon interaction.
- q_C : Probability that **cancer dies** upon T-cell interaction.
- $e^{-c_n C}$: Immune **down-regulation**; as cancer increases, ability to fight back decreases.

Simulating CML Dynamics (continuous imatinib)



Simulating CML Dynamics (with STI)



The Big Question

Motivated by ...

- *parameter value ranges* used in sensitivity analyses [Paquin *et al.*, 2011]
- patient recovery metrics: *time to remission*

Hematologic	Cytogenetic	Molecular	Total Elimination
$< 1.67 \text{ k}/\mu\text{L}$	$< 1.67 \cdot 10^{-2}$	$< 1.67 \cdot 10^{-4}$	$< 10^{-10}$

- the presence of *remission thresholds* over these ranges

Research Question

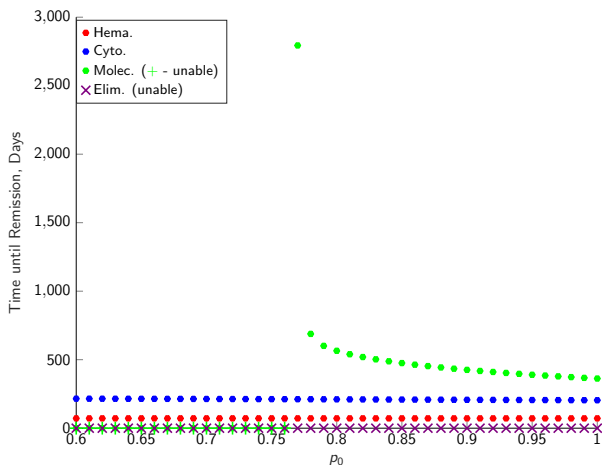
How does a treatment interruption alter parameter threshold values toward time to remission?

Note: no resistance case, i.e. $u = 0$.

Case Study: p_0 (Immune-Specific)

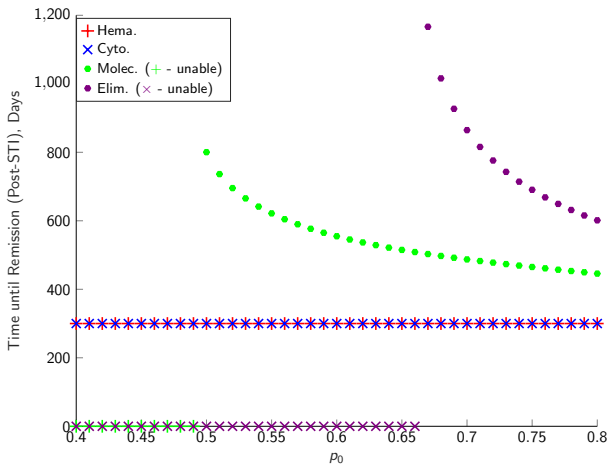
Consider p_0 , the probability that a T-cell encounters a cancer cell.
Under continuous imatinib treatment, remission thresholds at:

$p_0 \approx 0.77$ (Molecular); unable to completely eliminate CML.



Case Study: p_0

Incorporating a **15-day STI at 180 days**: remission thresholds at
 $p_0 \approx 0.5$ (Molecular), 0.67 (Elimination)



Conclusion and Next Steps

Key Takeaways:

- **Immune-specific** parameters: remission thresholds as *lower* bounds, further *decreased* with an STI.
- **CML-specific** parameters: remission thresholds as *upper* bounds, further *increased* with an STI.

Conclusion and Next Steps

Key Takeaways:

- **Immune-specific** parameters: remission thresholds as *lower* bounds, further *decreased* with an STI.
- **CML-specific** parameters: remission thresholds as *upper* bounds, further *increased* with an STI.

Future Work:

- Incorporating resistance: $u \neq 0$
- Finding “best” choice of treatment interruption plan (quantity, duration, frequency)
- Investigating “intermediate thresholds”
- Analytic insight: linearizing the DDE system, stability analysis